

Week 2 Introduction

Mean, Variance and Standard Deviation



Video: 2.1 – Introduction to Mathematical Concepts of Data Clustering
1 min



Practice Quiz: Population vs Sample – Review Information
3 questions



Video: 2.2 – Mean of One Dimensional Lists
2 min



Reading: Population vs Sample, Bias
10 min



Practice Quiz: Mean of One Dimensional Lists – Review Information
1 question



Video: 2.3 – Variance and Standard Deviation
3 min



Reading: Variability, Standard Deviation and Bias
10 min



Practice Quiz: Variance and Standard Deviation – Review Information
1 question

Development

Environment: Jupyter Notebooks

Computing Basic Statistics in Python

Week 2 Outro and Assessment



Population vs Sample, Bias

As we get into the meaty part of the mathematical underpinnings of the course, here are a few takeaways to keep in mind as you keep learning.

Population vs Sample

The core idea is the following: in practice, you almost never have access to the entire **population**, either because the data is unavailable, or because there are simply too many elements for you to study. Hence, you only take a portion, a **sample**, that you can study, and from which you can **estimate** what the total population looks like.

Remember the difference between the two situations we encounter:

- when we study the whole **population** (rarely in practice), the Greek letter μ is used to denote the (population) mean, and N (capital n) is used in the division.
- however, when we study only a **sample**, in order to make estimates about the population, $\bar{x}$ is the name used for the (sample) mean, we divide by n .

What is a bias (and an unbiased estimate)?

The sample mean is said to be an unbiased estimate of the population mean. What this means is that it is a good estimate of the population mean and can be proven mathematically. As Betty said, going deeper into the notion of a bias is beyond the scope of this MOOC. However, that does not mean you should not go and learn about it elsewhere (for instance, on [Wikipedia](#))!

The main idea here is the following: when taking a sample from a population, and then trying to figure out what the whole population looks like, it is more than likely that your sample actually does not reflect the entire population well! Take an oversimplified example: you have a population of three balls, and you sample only one from the three. Your three balls have all different colours, red, black and blue. If you only know the colour of your sample ball, let's say the blue one, it will obviously be biased in favour of that colour! In fact, it is so biased that you might think that all balls in your population are blue. You can now think of more general examples: a population of 100 people, the data in question being the size of